

When Skip Connections Bypass Prototypes: A Mechanistic Analysis of the Dice–Interpretability Trade-off in Cardiac Segmentation

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Abstract. U-Net-style prototype-based segmentation networks typically remove the skip connection, which reduces the Dice score. However, in medical settings, like cardiac segmentation, accurate segmentation is required. We present a mechanistic analysis of the Dice–interpretability trade-off using a 2×2 ablation study on MM-WHS CT data, crossing decoder architecture (with/without skip connections) and prototype scales. After verifying adding skip connections raises Dice but decreases average precision and prototype purity, three mechanistic analyses identify gradient bypass and poor spatial localisation as the two failure modes. Possible reasons and future work are discussed, including implications for trustworthy AI in clinical deployment.

Keywords: Prototype network · Cardiac Segmentation · Mechanistic Analysis.

1 Introduction

1.1 Motivation

Unlike post-hoc explanation, prototype networks offer structural interpretability through learned heatmaps, which is critical for trustworthy AI in high-stakes applications like cardiac segmentation [3]. However, a dilemma exists between interpretability and accuracy. Current prototype-based models like ProtoSeg [10] omit skip connections to enforce interpretability, but this sacrifices the spatial precision essential for biomedical structures [4]. In cardiac segmentation, where high Dice scores are clinically required, our experiments show that removing skip connections causes performance to collapse (Dice ≈ 0.60), making their inclusion necessary for deployment.

However, adding skip connections poses a risk of bypass effect. In a skip-connected decoder whose logits = $f(\mathbf{f}_{\text{enc}}, \mathbf{h}_{\text{proto}})$, the network may exploit low-resistance skip pathways, making prototypes causally irrelevant. We currently lack empirical evidence on whether this bypass is a causal failure (gradient disconnection) or a spatial one (misalignment). In this study, we conduct mechanistic analysis to understand the mechanism behind the bypass, thus providing possible mitigation.

1.2 Our Contributions

1. A 2×2 ablation experiment crossing architecture (with skip / without skip) and prototype level ($32 \times 32 + 16 \times 16$ / 16×16 only), producing a complete cross-factorial XAI characterisation of prototype segmentation models.
2. Mechanistic analysis identifying two distinct failure modes in skip-connected prototype segmentation: gradient bypassing and poor localisation. Additionally, single-level and multi-level prototype lead to different type of failure.

2 Related Work

2.1 Prototype networks

ProtoPNet [3] is an interpretable neural network architecture, which can explain predictions by comparison to a learned dictionary of visual prototypes. ProtoSeg [10] adapts ProtoPNet to semantic segmentation by attaching a prototype layer to an encoder-decoder backbone, introducing Jeffrey’s Diversity Loss to prevent intra-class prototype collapse, and projecting prototypes onto nearest real training patches. The present work is built on ProtoSeg, challenging whether its prototype heatmaps are causally valid.

[8] demonstrates in ScaleProtoSeg that different feature-map resolutions capture complementary semantic information and proposes scale-specific sparse prototype grouping, motivating us to use prototypes at multiple encoder levels.

2.2 Segmentation architectures and skip connections

The U-Net [9] introduces the encoder-decoder with skip connections as the standard architecture for biomedical image segmentation. [4] demonstrate that skip connections are important for recovering spatial detail in biomedical image segmentation. However, if a prototype layer and skip connections co-exist, it tends to be bypassed, which is an instance of shortcut learning [5]: networks preferentially exploit the lowest-loss signal pathway, bypassing the prototype bottleneck when skip-connected decoder routes provide a lower-resistance gradient path.

2.3 Mechanistic analysis of gradient routing and causal disconnection

Standard saliency maps often fail sanity checks [1], fostering genuine interpretability methods. From a mechanistic perspective, a model’s gradient flow should provide more reliable interpretability, which is helpful to identify shortcut learning. Thus, observing the ratio of gradient flow toward the decoder is an essential way to see if prototypes are bypassed.

To assess the spatial validity, researchers verify whether feature activations align with anatomical structures [7,11]. These methods are further supported by the concepts of counterfactual reasoning [6], which allow for the systematic isolation of failure modes. By comparing learned representations against ground-truth-centered oracles, researchers can distinguish whether localisation failures arise from a lack of encoder capacity or from a failure of the training objective.

3 Methods

3.1 Dataset

We used the CT volume from MM-WHS [2], split it into 16 training, 2 validation, and 2 test patients, yielding 3,389 / 382 / 484 2D axial slices, resampled to 256×256 size.

3.2 Model Architecture

Architectures with/without skip connections The skip-connected model has a hierarchical encoder (4 levels, channels [32, 64, 128, 256], output resolutions $128 \times 128 / 64 \times 64 / 32 \times 32 / 16 \times 16$) attaches prototype layers at selected levels (Table 1); Similarity heatmaps modulate encoder features as spatial attention masks before being forwarded to a standard U-Net decoder via skip connections at all levels.

For models without skip connections, the decoder is removed. Logits are a uniform average of upsampled prototype heatmaps across the selected levels $\mathcal{L}$: $\text{logits} = \frac{1}{|\mathcal{L}|} \sum_{l \in \mathcal{L}} \text{upsample}_{256}(\mathbf{H}^{(l)})$.

3.3 Experimental Design

The central experiment is a 2×2 factorial design crossing decoder architecture (with skip / without skip) against prototype levels:

1. **skip-16**: skip-connected with a 16×16 prototype level.
2. **skip-32+16**: skip-connected with $32 \times 32 + 16 \times 16$ prototype levels.
3. **noskip-16**: no skip connection with a 16×16 prototype level.
4. **noskip-32+16**: no skip connection with $32 \times 32 + 16 \times 16$ prototype levels.

Prototypes levels were chosen by a prior experiment. Table 1 shows that prototypes with 32×32 and 16×16 resolution provide better purity and Dice. L4 (16×16) captures the most semantically compressed representation; L3 (32×32) encodes finer-grained boundary texture. Thus, we used L4 for the single-scale prototype experiments, and L3+L4 for the multi-scale ones.

Prototype layer At level l , encoder features $\mathbf{F}^{(l)} \in \mathbb{R}^{C_l \times H_l \times W_l}$ are compared against a dictionary of prototypes $\{\mathbf{p}_{k,m}^{(l)}\} \subset \mathbb{R}^{C_l}$ ($K = 8$ classes). Similarity $s_{k,m}^{(l)} = \exp(-\|\mathbf{F}^{(l)} - \mathbf{p}_{k,m}^{(l)}\|^2)$ is aggregated via max-pooling into heatmaps $H_k^{(l)} = \max_m s_{k,m}^{(l)}$. A soft mask module then modulates features as $\tilde{\mathbf{F}}^{(l)} = \mathbf{F}^{(l)} \odot \mathbf{H}^{(l)}$, with final logits computed as $\sum_{l \in \mathcal{L}} w_l \cdot \text{upsample}_{256}(\mathbf{H}^{(l)})$. Joint training employs $\mathcal{L} = 0.5\mathcal{L}_{\text{Dice}} + 0.5\mathcal{L}_{\text{CE}} + \lambda_{\text{div}}\mathcal{L}_{\text{div}}$, where Jeffrey’s Divergence ($\mathcal{L}_{\text{div}}$) prevents prototype collapse. Finally, each $\mathbf{p}_{k,m}^{(l)}$ is grounded by projection to the nearest real training patch feature.

Table 1. Prototype levels, purity, and validation Dice scores.

Level	Spatial res.	Prototypes per class (M_l)	Purity	Val Dice (noskip)
1	128×128	4	0.159	0.146
2	64×64	3	0.569	0.336
3*	32×32	2	0.844	0.554
4*	16×16	2	0.679	0.606

*Selected for experiments.

3.4 Model training

All four models were trained with AdamW optimiser, learning rate 3×10^{-4} , weight decay 10^{-5} , batch size 16, and with cosine annealing over a maximum of 100 epochs. Early stopping was applied with patience 15 on validation Dice.

Epochs 1–20 optimised only the segmentation loss ($0.5 \mathcal{L}_{\text{Dice}} + 0.5 \mathcal{L}_{\text{CE}}$), allowing the encoder and prototype vectors to initialise before diversity is enforced. From epoch 21 onward, Jeffrey’s Divergence Loss ($\lambda_{\text{div}} = 0.01$) was added and prototype projection was applied every 10 epochs: each $\mathbf{p}_{k,m}^{(l)}$ was replaced by its nearest class- k training patch feature.

Training skip-32+16 from random initialisation caused prototype collapse at level 3. It was instead initialised from a fully converged skip-16 checkpoint, then fine-tuned for 100 epochs with both level 3 and level 4 prototype layers active.

3.5 Evaluation metrics

Average Precision (AP) For each foreground class k , we computed the prototype heatmap H_k (max-pooled across prototypes, upsampled to 256×256), and thresholded at the 95th percentile to produce binary mask M_k : $AP_k = \frac{|M_k \cap G_k|}{|M_k|}$. High AP indicates that regions of strong prototype activation correspond to the correct anatomical structure.

Prototype Purity This metric measures the fraction of each prototype’s nearest training patch that belongs to the represented class. Formally, let $\hat{y}(\mathbf{p}_{k,m}^{(l)})$ denote the ground-truth label at the spatial location of prototype $\mathbf{p}_{k,m}^{(l)}$ ’s nearest training patch; then:

$$\text{Purity}^{(l)} = \frac{1}{K \cdot M_l} \sum_{k=1}^K \sum_{m=1}^{M_l} \mathbf{1}[\hat{y}(\mathbf{p}_{k,m}^{(l)}) = k]$$

Multi-level Purity and AP For multi-level models, AP and Purity are weighted averages across levels, where each level’s weight ϕ_l is the fraction of test pixels at which that level produces the highest heatmap activation:

$$\text{Eff. AP} = \sum_{l \in \mathcal{L}} \phi_l \cdot \overline{AP}^{(l)}, \quad \text{Eff. Purity} = \sum_{l \in \mathcal{L}} \phi_l \cdot \text{Purity}^{(l)}$$

where $\overline{AP}^{(l)}$ is the mean AP over foreground classes at level l . For single-level models, Eff. AP and Eff. Purity reduce to the level’s own values.

3.6 Mechanistic Analysis

Gradient Attribution (bypass ratio) We define the bypass ratio as:

$$\text{bypass ratio} = \frac{\left\| \frac{\partial \text{logits}}{\partial \mathbf{f}_{\text{enc}}} \right\|}{\left\| \frac{\partial \text{logits}}{\partial \mathbf{f}_{\text{enc}}} \right\| + \left\| \frac{\partial \text{logits}}{\partial \mathbf{h}_{\text{proto}}} \right\|}$$

where $\mathbf{f}_{\text{enc}}$ is the encoder feature at level l and $\mathbf{h}_{\text{proto}}$ is the prototype-derived spatial activation at the same level. bypass ratio $\rightarrow 1$ indicates decoder gradient flows almost entirely through skip/encoder features; bypass ratio $\rightarrow 0$ indicates it flows through prototype heatmaps.

Spatial Precision For each model and foreground class k :

$$\text{SP}_k = \frac{\sum_{x,y} H_k(x,y) \cdot \mathbf{1}[G_k(x,y) = 1]}{\sum_{x,y} H_k(x,y)}$$

This quantity represents the fraction of heatmap activation mass falling on the correct ground-truth structure.

GT-Guided Counterfactual We construct an upper bound on heatmap localisation within each model’s own feature space. For each test slice at encoder level l , we compute the class centroid of the encoder features $\mathbf{F}_l$ under the downsampled GT mask, then build a Gaussian heatmap: $H_k^{\text{GT}}(i,j) = \exp(-|\mathbf{F}_l(:,i,j) - \mathbf{p}_k|^2/\sigma^2)$. Pixel-level AP is computed for both learned prototype heatmaps ($\text{AP}_{\text{learned}}$) and GT-guided heatmaps (AP_{GT}). The gap measures how far learned prototypes fall from the best localization the model’s own feature space could support.

4 Results

4.1 Dice-Interpretability trade-off

2 x 2 Ablation Table 2 reveals the same pattern in both single-level and multi-level model with skip connection: Dice is compatible with the baseline, but effective purity and effective AP drop significantly.

4.2 Mechanistic analysis

Gradient Attribution We examined whether the decrease of AP in skip models is due to the gradient flow bypassing prototypes. The results show that for the single-level (skip-16) model, 78% of gradient bypasses the prototypes, which echoes the assumptions. However, the multi-level (skip-32+16) model has more than 50% gradients that still go through the prototypes. The reason might be that the bypass ratio captures whether the output is causally sensitive to heatmap changes, while AP captures where the heatmap activates relative to the correct structure. Therefore, further mechanistic analysis for spatial precision is needed.

Spatial Precision Table 3 shows that skip-model heatmaps have only 26–32% of the spatial precision of no-skip heatmaps. This confirms that the bypass mechanism in skip models is not only causal (gradient bypass) but also spatial: prototypes activate at incorrect locations, producing heatmaps that do not represent the segmented structure.

GT-Guided Counterfactual Table 4 shows that skip-16 has learned heatmaps reaching 89% of the achievable upper bound, confirming that skip-16’s prototypes utilise localisation capacity. In contrast, skip-32+16’s prototype training produces poorly-localised heatmaps even relative to what its own encoder features could achieve.

5 Conclusion and Discussion

5.1 Mechanistic analysis summary

Our mechanistic analysis reveals that the dice-interpretability trade-off can have different causes: skip-16’s 78% of the gradient bypasses, thus the spatial precision is low; however, GT-guided counterfactual shows that the 22% residual gradient is enough to push prototype training close to its localisation ceiling. For the multi-level skip-32+16, gradient bypass is subtle, but its poor spatial precision can be explained by GT-guided counterfactual, which indicates that the 47%

Table 2. Main Results: 2×2 ablation study on architecture and prototype levels, compared with the baseline U-Net. All metrics are evaluated on the MM-WHS CT test set.

Model	Skip Conn.	Proto. Levels	Val Dice	Eff. AP	Eff. Purity
Baseline U-Net	Yes	None	0.823	0.349*	-
skip-16	Yes	16×16	0.810	0.057	0.474
noskip-16	No	16×16	0.606	0.312	0.679
skip-32+16	Yes	32×32+16×16	0.821	0.051	0.527
noskip-32+16	No	32×32+16×16	0.559	0.301	0.686

*For U-Net, the final softmax segmentation is used as a proxy for AP evaluation.

Table 3. Spatial Precision (SP) comparison between no-skip and skip variants. The numbers in parentheses are the percentage of skip to no-skip.

Resolution	No-Skip SP	Skip SP
16×16	0.075	0.020 (26.7%)
$32\times 32+16\times 16$	0.062	0.020 (32.3%)

residual gradient is insufficient to drive prototype localisation. The multi-level model’s different prototype failure can be attributed to several reasons:

1. In max-aggregation, the 32×32 layer dominates 60% of pixels. Each cell of this layer spans 8×8 pixels, where the activation pattern is broad coverage of large anatomical regions rather than precise boundary localisation. The 53% of gradients that reach prototypes therefore mostly rewards 32×32 prototypes for wide spatial coverage, not accurate localisation.
2. The GT-guided upper bound is lower (0.389 vs. 0.623), which is counterintuitive given that more levels are available. The likely cause is max-aggregation: 32×32 ’s higher activation magnitude over large regions dominates the combined heatmap, suppressing 16×16 ’s sharper activations.
3. Skip-32+16 has bypass paths from two layers; although bypass ratio is lower, the bypass can be spatially more complete.
4. Two prototype levels compete for gradient, thus both are undertrained relative to the single-level model.

5.2 Is the trade-off fundamental?

The skip–no-skip trade-off is consistent across both 16×16 -only and $32\times 32+16\times 16$ configurations, and in both directions: removing skip connections raises AP; restoring them suppresses it. This implies the trade-off is structural rather than tunable through further training. Prototype training improvements (diversity losses, push-pull regularisation, attention clustering) may increase prototype vector quality, but they cannot force the decoder to use the prototype signal as long as the skip pathway remains available. From a trustworthy AI perspective, this implies that interpretability claims for skip-connected prototype models should be empirically validated rather than assumed, as the structural bypass identified here can silently invalidate heatmap-based explanations.

5.3 Future Direction

1. **Progressive skip ablation:** Monotonically remove skip connections ($16\times 16 \rightarrow 32\times 32+16\times 16 \rightarrow \text{all}$) to assess whether AP degrades monotonically.

Table 4. Gap between GT upper bound and learned AP.

Model	AP_{GT} (upper bound)	$\text{AP}_{\text{learned}}$	$\text{AP}_{\text{GT}} - \text{AP}_{\text{learned}}$
skip-16	0.623	0.551	0.072 (-11%)
skip-32+16	0.389	0.078	0.311 (-80%)

2. **Heatmap-regularised decoder:** Train a skip decoder with an explicit heatmap alignment loss that penalise outputs that are insensitive to heatmap substitution. This would combine high Dice with learned bypass resistance.
3. **Spatial attention pre-filtering:** Apply a learned spatial mask prior to prototype matching to improve prototype localisation (spatial precision) without architectural changes.
4. **MRI modality generalisation:** All results are CT-only. The skip-interpretability trade-off may differ for MRI, where texture statistics and class boundaries differ substantially.

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